UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
COURSE CODE	:	**CSC429**
EXAMINATION	:	**JULY 2023**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (10 Questions)
   PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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PART A

QUESTION 1

A touch screen is an essential device commonly used to ensure work can be conducted with ease and convenience. It can act as input and output devices

a) Describe how the touch screen can be an input and output device.

(4 marks)

b) Name **ONE (1)** another device that can be an input and output.

(1 mark)

QUESTION 2

John Von Neumann is a mathematician who is one of the pioneers in computer advancement, and his contributions are still relevant to the modern computer.

a) Draw the Von Neuman architecture and give relevant examples each.

(4 marks)

b) What is the name of the actual device that John Von Neumann himself developed?

(1 mark)

QUESTION 3

Solve the following equations. Show the complete workflow.

a) $52.73_8 + AB.2_H =$ _____$_2$.

(2 marks)

b) $171.28125_{10} + 1010\ 1100.11_2 =$ _____ $_8$.

(3 marks)

QUESTION 4

a) Solve $-48_{10} + 68_{10}$ using the sign and magnitude method. Show the complete workflow.

(2 marks)

b) Calculate $-96_{10} -15_{10}$ using the 1's and 2's complement method. Show the complete workflow.

(3 marks)

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QUESTION 5

a) Convert 273.78125_{10} to a floating-point number in IEEE single precision format in hexadecimal value form.

(5 marks)

b) Convert C158 8000_H in IEEE single precision floating point number to decimal value form.

(5 marks)

QUESTION 6

a) List the phases in the machine cycle.

(4 marks)

b) Discuss the function of the different CPU components in the machine cycle.

(4 marks)

c) What will happen in the machine cycle if access to the main memory is blocked?

(2 marks)

QUESTION 7

a) List **TWO (2)** buses types that are available in the computer system.

(2 marks)

b) Differentiate **TWO (2)** characteristics of system interconnection and CPU interconnection.

(4 marks)

c) Explain the different types of data transfer timing.

(4 marks)

QUESTION 8

a) Discuss the main function of cache memory.

(2 marks)

b) How can the data stored in the cache and main memory be synchronized.

(4 marks)

c) Contrast between SRAM and DRAM based on the following characteristics:

i. Type of storage
ii. Availability of charges leak
iii. Price
iv. Example of location of memory usage

(4 marks)

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QUESTION 9

Question 9 is based on Figure 1. Figure 1 illustrates the multiple interrupts that occur in a computer system.

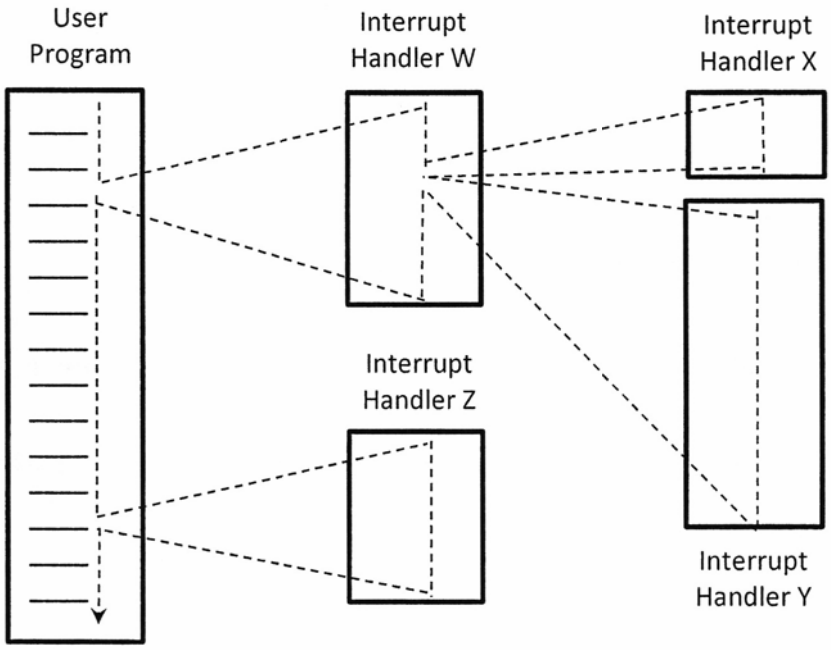


Figure 1: Multiple interrupts

a) Define interrupt.

(2 marks)

b) Rank the interrupts based on their priority from the most important to the least important.

(4 marks)

c) Rank the interrupts from the most important to the least important if disable interrupt is applied.

(4 marks)

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QUESTION 10

Question 10 is based on Figure 2. Figure 2 depicts a Non-uniform Memory Access (NUMA) architecture.

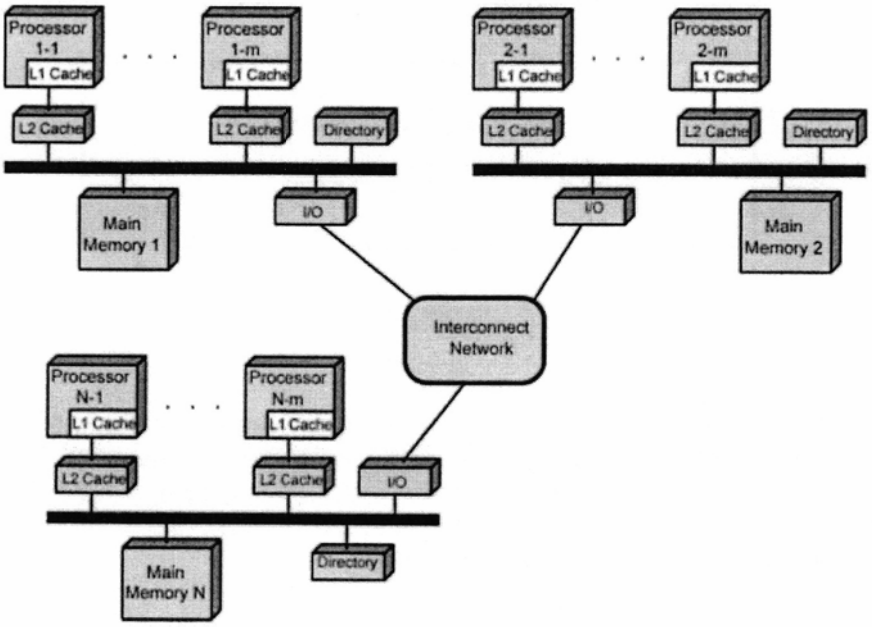


Figure 2: Non-uniform Memory Access (NUMA)

a) Redraw a symmetric multiprocessor organization based on Figure 2.

(2 marks)

b) Give **TWO (2)** examples of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

(4 marks)

c) Discuss **TWO (2)** disadvantages of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

(4 marks)

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PART B

QUESTION 1

Question 1 is based on Figure 3 and Table 1.

$$y = \frac{A^2 - (B + AC)}{AD}$$

Figure 3

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 3 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Note that Register ***A***, ***B, C,*** and ***D*** cannot be modified, and Register ***x, y***, and ***z*** can be used as temporary registers. However, the final answer must be stored in Register ***y***.

(20 marks)

Table 1: Machine Instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	

END OF QUESTION PAPER

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